

ENDG 319

Probability, Statistics and Machine Learning

Lecture 2: Numerical Summaries of Data

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From sample data to useful information

A sample may contain dozens, hundreds, or millions of observations.



Good summaries reduce complexity without hiding the important structure.

Learning outcomes for Lecture 2

BY THE END OF TODAY, STUDENTS SHOULD BE ABLE TO

- Compute and interpret the sample mean, variance, and standard deviation.
- Explain how outliers affect the mean and standard deviation.
- Compute and interpret the median, quartiles, percentiles, range, and IQR.
- Summarize categorical data using frequencies and sample proportions.
- Distinguish sample statistics from population parameters.

Core skill: choose a summary that matches the type and shape of the data.

A motivating dataset: particulate-matter emissions

Does altitude affect particulate-matter emissions from vehicles?

LOW ALTITUDE • n = 138

TABLE 1.1 Particulate matter (PM) emissions (in g/gal) for 138 vehicles driven at low altitude

1.50	0.87	1.12	1.25	3.46	1.11	1.12	0.88	1.29	0.94	0.64	1.31	2.49
1.48	1.06	1.11	2.15	0.86	1.81	1.47	1.24	1.63	2.14	6.64	4.04	2.48
2.98	7.39	2.66	11.00	4.57	4.38	0.87	1.10	1.11	0.61	1.46	0.97	0.90
1.40	1.37	1.81	1.14	1.63	3.67	0.55	2.67	2.63	3.03	1.23	1.04	1.63
3.12	2.37	2.12	2.68	1.17	3.34	3.79	1.28	2.10	6.55	1.18	3.06	0.48
0.25	0.53	3.36	3.47	2.74	1.88	5.94	4.24	3.52	3.59	3.10	3.33	4.58
6.73	7.82	4.59	5.12	5.67	4.07	4.01	2.72	3.24	5.79	3.59	3.48	2.96
5.30	3.93	3.52	2.96	3.12	1.07	5.30	5.16	7.74	5.41	3.40	4.97	11.23
9.30	6.50	4.62	5.45	4.93	6.05	5.82	10.19	3.62	2.67	2.75	8.92	9.93
6.96	5.78	9.14	10.63	8.23	6.83	5.60	5.41	6.70	5.93	4.51	9.04	7.71
7.21	4.67	4.49	4.63	2.80	2.16	2.97	3.90					

HIGH ALTITUDE • n = 62

TABLE 1.2 Particulate matter (PM) emissions (in g/gal) for 62 vehicles driven at high altitude

7.59	6.28	6.07	5.23	5.54	3.46	2.44	3.01	13.63	13.02	23.38	9.24	3.22
2.06	4.04	17.11	12.26	19.91	8.50	7.81	7.18	6.95	18.64	7.10	6.04	5.66
8.86	4.40	3.57	4.35	3.84	2.37	3.81	5.32	5.84	2.89	4.68	1.85	9.14
8.67	9.52	2.68	10.14	9.20	7.31	2.09	6.32	6.53	6.32	2.01	5.91	5.60
5.61	1.50	6.46	5.29	5.64	2.07	1.11	3.32	1.83	7.56			

Excerpts shown from the two samples in the 2024 lecture notes.

Raw data are difficult to compare by eye. We need a few interpretable summaries.

What do summary statistics tell us?

CENTRE

What is a typical value?

Mean, median

SPREAD

How much do observations vary?

Standard deviation, range, IQR

POSITION

Where does a value fall within the sample?

Quartiles, percentiles

CATEGORIES

How many observations fall in each group?

Frequencies, sample proportions

A statistic is a numerical characteristic calculated from a sample.

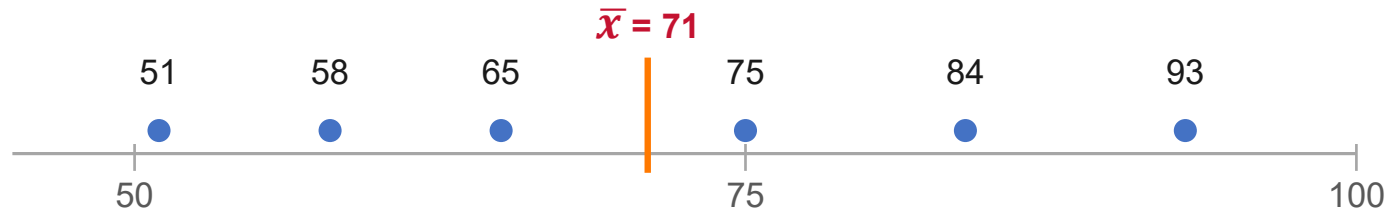


The sample mean: a measure of centre

Salaries of six engineers (in \$1000):

51, 58, 65, 75, 84, 93

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$



$$\bar{x} = \frac{51 + 58 + 65 + 75 + 84 + 93}{6} = 71$$

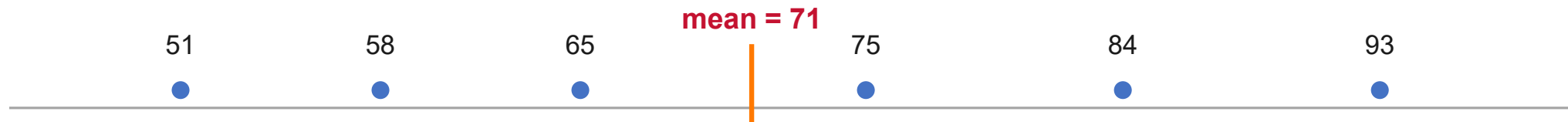
INTERPRETATION

The average salary is
71 thousand dollars
= \$71,000.

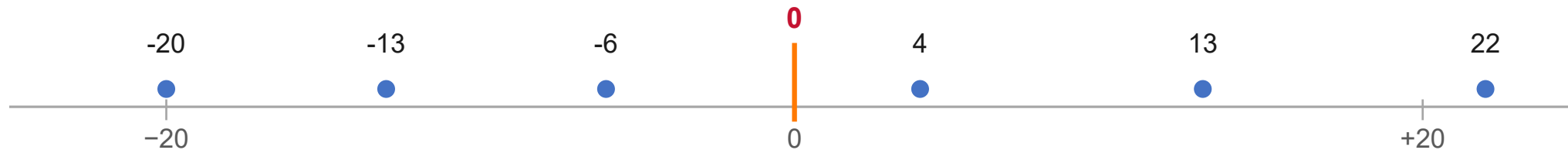
The mean is an arithmetic balance point—not necessarily one of the observed values.

Why the mean is a centre

Original observations



Deviations from the mean: $x_i - \bar{x}$



$$\sum_{i=1}^n (x_i - \bar{x}) = 0$$

Negative and positive deviations cancel exactly: the mean is the balance point.

Sample variance and standard deviation

DEVIATION

$$d_i = x_i - \bar{x}$$

Signed distance from the mean

SAMPLE VARIANCE

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

Average squared deviation

SAMPLE STANDARD DEVIATION

$$s = \sqrt{s^2}$$

Typical distance from the mean

Equivalent computational form:

$$s^2 = \frac{\sum_{i=1}^n x_i^2 - n\bar{x}^2}{n-1}$$

Units: $\bar{x}$ and s have the same units as the data; s^2 has squared units.

Computing s: the salary example

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
51	-20	400
58	-13	169
65	-6	36
75	+4	16
84	+13	169
93	+22	484
Sum = 426	Sum = 0	Sum = 1274

$$n = 6, \bar{x} = 71$$

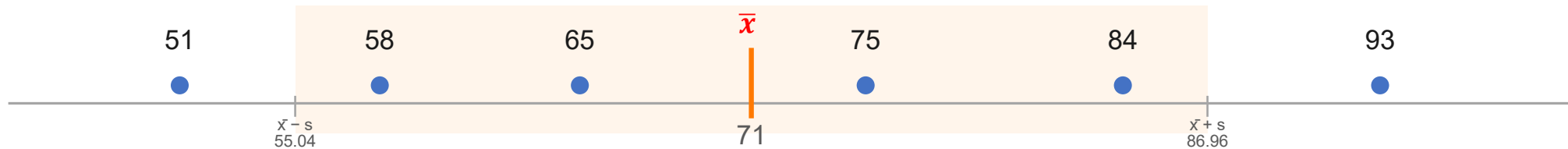
$$s^2 = \frac{1274}{6 - 1} = 254.8$$

$$s = \sqrt{254.8} = 15.96$$

Interpretation: salaries typically differ from the mean by about \$15.96 thousand.

Interpreting standard deviation

For the salary data: $\bar{x} = 71$ and $s = 15.96$



$$s \geq 0$$

$s = 0$ only when all values are identical

s has the same units as the data

Standard deviation describes spread; it does not guarantee a fixed percentage within $\bar{x} \pm s$.

Same mean, different spread

Dataset A — tightly clustered



Dataset B — moderate spread



Dataset C — wide spread



All three samples have $\bar{x} = 75$. The standard deviation distinguishes their variability.

Changing units: linear transformations

$$Y_i = a + bX_i \longrightarrow \bar{Y} = a + b\bar{X} \quad s_Y^2 = b^2 s_X^2 \quad s_Y = |b|s_X$$

SHIFT BY a

Adding a constant changes the mean by a, but leaves the standard deviation unchanged.

SCALE BY b

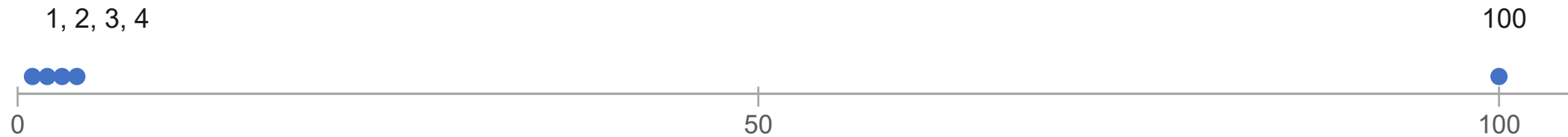
Multiplying every value by b multiplies the mean by b and the standard deviation by |b|.

Example: Celsius $\rightarrow$ Fahrenheit $F = 32 + 1.8C \longrightarrow \bar{F} = 32 + 1.8\bar{C}, \quad s_F = 1.8s_C$

A change of units changes the numerical summaries—but not the underlying information.

Outliers: investigate before acting

An outlier is an observation that is much larger or smaller than the rest of the sample.



Possible data-entry or measurement error

Genuine but unusual observation

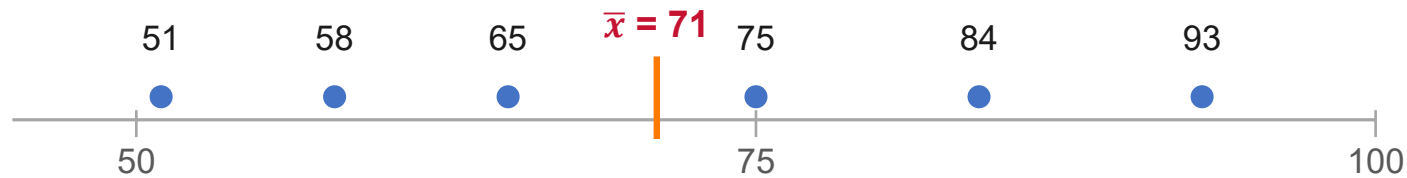
Process change, failure, or special cause

Observation from a different population

Verify → investigate context → compare analyses → document the decision. Do not delete automatically.

How an outlier changes the summaries

Six engineers



Mean = 71
Median = 70
SD = 15.96

Add CEO salary: 300



Mean = 103.71
Median = 75
SD = 87.77

The mean and SD are sensitive to extreme values; the median is resistant.

The sample median: a resistant measure of centre

The median divides the ordered sample into two approximately equal parts.

ODD n

$$\text{Median position} = \frac{n + 1}{2}$$

Use the single observation in that position.

EVEN n

$$\text{Median} = \frac{x_{(n/2)} + x_{(n/2+1)}}{2}$$

Average the two middle observations.

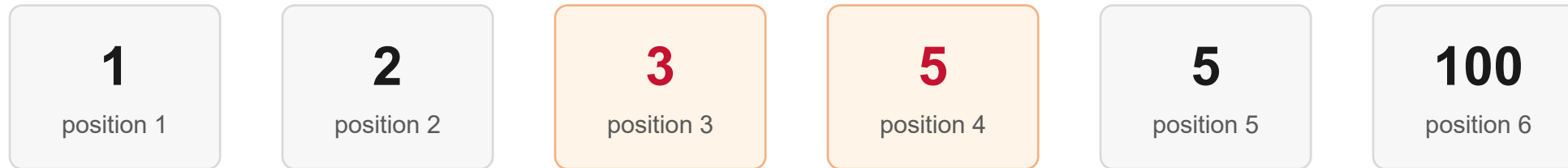
Always order the data from smallest to largest first.

Because the median uses ranks, one very large or small observation has limited influence.

Median example

Sample: 1, 2, 5, 3, 5, 100

→ Sort the data



$$\text{Median} = \frac{3 + 5}{2} = 4$$

For comparison:
Mean = 19.33

The extreme value 100 pulls the mean upward, but the median remains near the centre of most observations.

Mean, median, and trimmed mean

Sample	Mean	Median
1, 2, 3, 4	2.5	2.5
1, 2, 3, 4, 100	22.0	3.0
1, 2, 3, 4, 100, 1000	185.0	3.5

MEAN
Uses every value; sensitive to extremes.

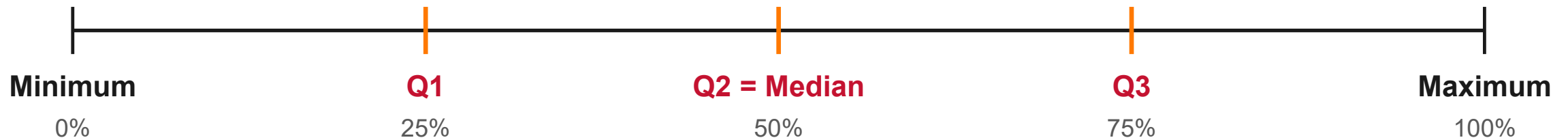
MEDIAN
Uses order; resistant to extremes.

10% trimmed mean Order the data, remove the smallest 10% and largest 10%, then average the remaining observations.

Choice of centre should reflect the shape of the data and the purpose of the analysis.

Quartiles and the five-number summary

Quartiles divide ordered data into four approximately equal parts.



INTERPRETATION

About 25% of values lie below Q1; about 50% below Q2; about 75% below Q3.

COURSE LOCATION RULE

$$L_{Q_1} = 0.25(n + 1) \quad L_{Q_2} = 0.50(n + 1)$$

$$L_{Q_3} = 0.75(n + 1)$$

Five-number summary = minimum, Q1, median, Q3, maximum.

Percentiles

The p th percentile separates approximately the lowest $p\%$ of the data from the highest $(100 - p)\%$.

$$L_p = \frac{p}{100} (n + 1)$$

If L_p is an integer

Use the observation in that ordered position.

If L_p is not an integer

Using the course convention, average the two observations on either side.

Special cases: 25th percentile = Q1; 50th = median; 75th = Q3. The 95th percentile leaves about 5% above it.

Software conventions can differ—use the method specified for this course.

In-class example

Consider the ordered sample:

	2	3	5	6	7	9	9	11	12	15
Position	1	2	3	4	5	6	7	8	9	10

1. What is the sample mean?
2. What is the median?
3. Using the course convention, what are Q1 and Q3?
4. What are the range and IQR?

Show your position calculations.

Worked solution

MEAN

$$\bar{x} = \frac{79}{10} = 7.9$$

MEDIAN

$$Q_2 = \frac{7 + 9}{2} = 8$$

FIRST QUARTILE

$$L_{Q_1} = 2.75 \Rightarrow Q_1 = \frac{3 + 5}{2} = 4$$

THIRD QUARTILE

$$L_{Q_3} = 8.25 \Rightarrow Q_3 = \frac{11 + 12}{2} = 11.5$$

Five-number summary

2

min



4

Q1



8

median



11.5

Q3



15

max

$$\text{Range} = 15 - 2 = 13$$

$$\text{IQR} = 11.5 - 4 = 7.5$$

Range and interquartile range (IQR)

Range = maximum – minimum

$$\text{IQR} = Q_3 - Q_1$$

Sample	Range	IQR
2, 3, 5, 6, 7, 9, 9, 11, 12, 15	13	7.5
2, 3, 5, 6, 7, 9, 9, 11, 12, 150	148	7.5

Range uses only the two extremes and is highly sensitive to outliers.

IQR describes the middle 50% and is resistant to extreme values.

Categorical data: frequency and sample proportion

FREQUENCY

f_j = number of sample items
in category j

SAMPLE PROPORTION / RELATIVE FREQUENCY

$$\hat{p}_j = \frac{f_j}{n}$$

Useful engineering categories

Pass / fail

Failure mode

Material grade

Machine line

Checks: $\sum f_j = n$ and $\sum \hat{p}_j = 1$.

Rivet example

A sample of 100 rivets is checked for breaking strength. Four rivets fail.

Outcome	Frequency	Sample proportion
Pass	96	0.96
Fail	4	0.04

Failure proportion

$$\hat{p}_{\text{fail}} = \frac{4}{100} = 0.04 \\ = 4\%$$

Correct interpretation: 4% of the sampled rivets failed.

Inference is required before generalizing to all rivets produced.

Sample statistics and population parameters

Quantity	Sample statistic	Population parameter
Mean	$\bar{x}$	μ
Variance	s^2	σ^2
Standard deviation	s	σ
Proportion	$\hat{p}$	p



**STATISTICS
ESTIMATE
PARAMETERS**

Observed from the sample

Usually unknown for the population

Key takeaways

- The sample mean describes centre; the sample standard deviation describes spread.
- Mean and standard deviation are sensitive to extreme values.
- Median and IQR are resistant summaries based on ordered positions.
- Quartiles and percentiles describe where observations lie within a sample.
- Frequencies and sample proportions summarize categorical data.
- Sample statistics are used to estimate population parameters.

Next: graphical summaries and the shape of a distribution.